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Small Bank Lending in the Era of Fintech and Shadow Banks: A Sideshow?

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1 Big picture

- **Question.** In the post-2008 mortgage market, why did *small banks* remain relevant even as nonbanks, shadow banks, and fintech lenders rapidly expanded and the rest of traditional banking lost market share?
- **Setting.** The paper studies the U.S. mortgage origination market from 2008 to 2015, focusing on cross-county differences in lender growth across seven lender classes: the Big4, large banks, intermediate banks, small banks, credit unions, shadow banks, and fintech lenders.
- **Core challenge.** Aggregate market-share facts are not enough to distinguish between a broad secular rise of nonbanks and a local supply-side substitution in which some lenders step in where others retreat. The paper therefore needs a way to isolate *local supply shocks* from changes in local demand.
- **Main result.** Small banks did not simply survive as a residual category. They substituted strongly for the retreating Big4 in local mortgage markets, and this response was much larger than the local response of shadow banks or fintech lenders.
- **Mechanism 1: differential regulation.** Postcrisis regulatory burden—capital pressure, mortgage-servicing-right rules, and lawsuits—rose with bank size and hit the Big4 especially hard. Small banks faced a lighter regulatory touch and exploited the resulting opening.
- **Mechanism 2: GSE financing cost changes.** Changes in GSE guarantee fees between 2011 and 2015 removed volume discounts for the largest GSE sellers and raised overall GSE funding costs. These shocks opened room for small banks both in GSE lending and in balance-sheet lending.
- **Key comparative advantage.** Relative to nonbanks, small banks can lend on balance sheet. This makes them especially effective at replacing the Big4 when retained lending becomes relatively more valuable.
- **Main quantitative facts.** Small banks’ aggregate mortgage share rose from 16% in 2008 to 19% in 2015, while the Big4 fell from 33% to 12%. In county-level IV regressions, a 10-percentage-point larger decline in Big4 growth corresponds to about an 8.8–9.7 percentage point increase in small-bank growth, versus much smaller responses from shadow banks and fintech.
- **Why this paper matters.** The paper shows that “traditional banks” should not be treated as a homogeneous block. Within-bank-size heterogeneity matters for understanding postcrisis credit supply, market structure, and the channels through which regulation reshapes lending.

2 Data and measurement (key objects)

2.1 Observed lender-county mortgage panel

- **Main data source.** The core data are mortgage originations from the Home Mortgage Disclosure Act (HMDA) Loan Application Register.
- **Coverage.** The analysis keeps originated loans whose purpose is either home purchase or refinancing.

- **Level of aggregation.** The main empirical object is a **lender-class** $\times$ **county** panel, mainly over 2008–2015, with some exercises focusing on 2011–2015.
- **Parent-level lender mapping.** Because HMDA respondents report direct parents rather than ultimate holding companies, the paper uses the HMDA Lender File from Robert Avery to map depository institutions and mortgage companies to their ultimate parents. This avoids misclassifying affiliated mortgage companies as stand-alone lenders.
- **Lender classes.** The seven classes are:
 - **Big4:** Bank of America, Citi, JP Morgan, Wells Fargo,
 - **Large banks:** assets between \$50 billion and \$1 trillion,
 - **Intermediate banks:** assets between \$10 billion and \$50 billion,
 - **Small banks:** assets below \$10 billion,
 - **Credit unions,**
 - **Shadow banks:** nonbank, non-credit-union, non-fintech lenders,
 - **Fintech lenders:** nonbanks with significant online presence and minimal human interaction in the application process, following Buchak et al.
- **Growth outcome.** For lender class LC in county c , the main growth measure is

$$\text{Growth}_c^{LC} = 100 \times \frac{\text{Originations}_{c,2015}^{LC} - \text{Originations}_{c,2008}^{LC}}{\text{Originations}_{c,2008}^{All}}.$$

- **Interpretation.** The numerator is the lender-class-specific change in lending; the denominator scales it by the county’s initial overall lending. This makes the coefficient interpretable in percentage-point terms at the county level.

2.2 Loan segments and market segments

- **Balance-sheet lending.** Loans are treated as on-balance-sheet if HMDA classifies them as “not sold” or “sold to an affiliate.”
- **GSE lending.** Loans sold to Fannie Mae or Freddie Mac are classified as GSE loans.
- **Jumbo loans.** Loans above the conforming loan limit are used as a placebo segment when the paper studies GSE guarantee-fee shocks, because those loans cannot be sold to the GSEs.
- **Why these segments matter.** They line up directly with the paper’s two main mechanisms:
 - regulation and capital pressure should matter especially for **balance-sheet** lending,
 - changes in GSE guarantee fees should matter especially for **GSE** lending and for retained lending that becomes relatively more attractive when securitization costs rise.

2.3 Observed county controls

- **Demographics and local conditions.** The controls include 2008 county-level population, minority share, per-capita income, median age, subprime share (share with FICO below 660),

house-price growth, banking concentration measured by deposit HHI, the number of banks with a physical branch, the number of banks with mortgage activity, and the number of nonbank lenders with mortgage activity.

- **Why these controls matter.** The identification challenge is that a lender’s retreat from a county might reflect changing local demand, not just supply. These controls are meant to absorb the most obvious sources of demand heterogeneity.
- **Summary facts.** The average county in 2008 had population 89.24 thousand, minority share 0.21, income \$33.65 thousand, subprime share 0.33, median age 39.39, and house-price growth over 2008–2015 of -2.14% . The average county banking HHI was 0.30.

2.4 Observed lender shares and initial conditions

- The average 2008 county lending shares were approximately:

Big4 27.1%, Large banks 12.1%, Intermediate banks 8.9%,

Small banks 30.1%, Credit unions 5.2%, Shadow banks 14.9%, Fintech 1.6%.

- **Interpretation.** Small banks started with a large local presence in many counties even though their aggregate national share was modest. This is one reason why local substitution can matter even when aggregate market-share changes look small.

2.5 Shock measures

- **Regulatory shocks.** The paper measures postcrisis regulatory burden using changes in tier 1 capital ratios, mortgage servicing rights (MSR) relative to capital, and mortgage-related lawsuits/fines.
- **GSE funding shocks.** The paper measures exposure to changes in guarantee fees by ranking lenders by the volume of loans sold to the GSEs in 2011 and constructing county-level exposure to top GSE sellers.
- **Top-10 GSE exposure.** The county-level exposure variable is

$$\text{Top10GSELenderShare}_{c,2011} = \frac{\text{Originations}_{c,2011}^{\text{Top10 GSE lenders}}}{\text{Originations}_{c,2011}^{\text{All}}}.$$

- **Interpretation.** Counties more exposed to the top GSE sellers are more exposed to the removal of high-volume GSE fee discounts and thus more exposed to the parity shock.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline local substitution regression

The first empirical specification studies how each lender class responds to county-level Big4 retreat:

$$\text{Growth}_c^{\text{LenderClass}} = \phi \text{Growth}_c^{\text{Big4}} + \zeta_{\text{state}(c)} + X_c' \beta + \varepsilon_c. \quad (1)$$

- **Interpretation.** The coefficient ϕ measures whether a lender class grows more where the Big4 shrink more.
- **What the coefficient means.** A negative ϕ implies local substitution: when Big4 growth is lower, that lender class grows more.
- **Why the paper starts here.** This is the simplest way to uncover cross-county heterogeneity in substitution patterns before moving to an IV design.

3.2 Main result from the OLS specification

In the OLS regressions, the estimated coefficients are:

$$\hat{\phi}_{\text{small banks}} = -0.659, \quad \hat{\phi}_{\text{shadow banks}} = -0.130, \quad \hat{\phi}_{\text{fintech}} = -0.047.$$

- **Interpretation.** A 10-percentage-point lower Big4 growth is associated with:
 - 6.6 percentage points higher small-bank growth,
 - 1.3 percentage points higher shadow-bank growth,
 - about 0.5 percentage points higher fintech growth.
- **Economic meaning.** The local response of small banks is much larger than the local response of nonbanks. This already suggests that the persistence of small banks is not just the mirror image of the national rise of shadow banks.

3.3 Why OLS is not enough

- County-level Big4 retreat could be correlated with local demand shifts rather than supply shocks. For example, counties might simultaneously experience a fall in demand for Big4-style mortgage products and a rise in demand for products more likely to be offered by small banks.
- This would bias equation (1), because $\text{Growth}_c^{\text{Big4}}$ would no longer be exogenous.
- The paper therefore turns to instruments designed to isolate exogenous variation in the *supply-side* scope for Big4 withdrawal.

3.4 Instrument 1: the 2008 Big4 share

The first-stage and second-stage IV equations are

$$\text{Growth}_c^{\text{Big4}} = \theta \text{Share08}_c^{\text{Big4}} + \zeta_{\text{state}(c)} + X_c' \beta + \eta_c, \quad (2)$$

$$\text{Growth}_c^{\text{LenderClass}} = \psi \widehat{\text{Growth}_c^{\text{Big4}}} + \xi_{\text{state}(c)} + X_c' \gamma + u_c. \quad (3)$$

- **Economic idea.** Counties where the Big4 had larger initial market share had more scope for meaningful withdrawal after the postcrisis regulatory shock.
- **First-stage result.** A 10-percentage-point higher Big4 share in 2008 is associated with a 6.9-percentage-point lower Big4 growth rate from 2008 to 2015.

- **Why it helps.** The instrument uses *predetermined exposure* rather than contemporaneous retreat. That helps separate local demand conditions from the Big4’s ability or need to reduce lending.

3.5 Instrument 2: exposure to mortgage-lawsuit penalties

The second IV uses county exposure to the postcrisis mortgage lawsuits that hit the Big4 disproportionately:

$$\text{Lawsuit}_c = 100 \times \sum_{b \in c} L_b \frac{\text{Originations}_{bc,2008}}{\sum_{d \in c} \text{Originations}_{dc,2008}}, \quad (4)$$

where L_b is the accumulated 2008–2015 lawsuit exposure for bank b .

- **Interpretation.** This instrument captures how much a county was exposed to a national bank-level regulatory shock that hit the Big4, especially the hardest-hit members of that group.
- **Why it is appealing.** The lawsuits were bank-level, nationwide, and concentrated on the Big4, while small banks were largely untouched.
- **First-stage strength.** The first-stage F-statistic is above 900 with the lawsuit instrument and above 2,000 with the initial-share instrument.

3.6 Main IV result

Using the two IVs, the estimated substitution effects are much stronger for small banks than for nonbanks:

$$\hat{\psi}_{\text{small banks}} = -0.884 \quad (\text{Big4-share IV}), \quad \hat{\psi}_{\text{small banks}} = -0.970 \quad (\text{lawsuit IV}).$$

- **Interpretation.** A 10-percentage-point larger decline in Big4 growth corresponds to about 8.8–9.7 percentage points higher small-bank growth.
- **Contrast with nonbanks.** The same 10-point Big4 decline corresponds to only about 2.6–3.5 points for shadow banks and roughly 1.2–1.7 points for fintech.
- **Conclusion.** The strongest local supply-side substitute for the retreating Big4 is the small-bank sector, not nonbanks.

3.7 Balance-sheet lending as the comparative-advantage channel

To compare balance-sheet and non-balance-sheet responses, the paper stacks county-level observations by loan segment and estimates

$$\text{Growth}_c^{\text{Big4}} = \gamma \text{Share08}_c^{\text{Big4}} + \text{controls} + \nu_c, \quad (5)$$

$$\text{Growth}_c^{\text{Big4}} \times D_c^{BS} = \theta \text{Share08}_c^{\text{Big4}} + \text{controls} + \eta_c, \quad (6)$$

$$\text{Growth}_c^{\text{LenderClass}} = \psi \text{Growth}_c^{\text{Big4}} + \phi (\text{Growth}_c^{\text{Big4}} \times D_c^{BS}) + \text{controls} + \varepsilon_c, \quad (7)$$

where $D_c^{BS} = 1$ for balance-sheet lending observations.

- **Interpretation.** ψ is the substitution effect in non-balance-sheet lending. ϕ is the extra response in balance-sheet lending.
- **Small-bank estimates.** For small banks, $\hat{\psi} = -0.567$ and $\hat{\phi} = -0.580$, so the overall balance-sheet response is

$$\hat{\psi} + \hat{\phi} = -1.147.$$

- **Economic interpretation.** Small-bank balance-sheet lending is about *twice as responsive* as their non-balance-sheet lending. This is the clearest evidence that their key comparative advantage over nonbanks is the ability to hold loans on their own balance sheet.
- **Why this matters.** Nonbanks can exploit lighter regulation, but they cannot replicate the bank business model of retained lending. That is why the local substitution effect of small banks can exceed that of shadow banks and fintech despite nonbanks' aggregate growth.

3.8 The GSE guarantee-fee parity shock

The second supply-side mechanism comes from changes in the cost of selling loans to Fannie Mae and Freddie Mac.

- **Institutional change.** Beginning in 2011, the FHFA required the GSEs to raise guarantee fees and to move toward *uniform pricing* across lenders, reducing or eliminating the volume discounts previously enjoyed by the largest GSE sellers.
- **Two parts of the shock.**
 - **Parity shock:** the removal of size-based discounts for large-volume sellers,
 - **Level shock:** the average level of guarantee fees roughly doubled from about 30 bps to about 60 bps.
- **Why this matters.** The parity shock leveled the playing field in the GSE segment for smaller lenders; the level shock raised the cost of securitization more generally and made loan retention relatively more attractive.

3.9 Parity-shock regression for small-bank GSE growth

The paper estimates

$$\text{GSEGrowth}_c^{\text{SmallBank}} = \pi \text{Top10GSELenderShare}_{c,2011} + \zeta_{\text{state}(c)} + X'_c \beta + \varepsilon_c. \quad (8)$$

- **Interpretation.** Counties that were more dominated by the top GSE sellers in 2011 were more exposed to the removal of large-lender discounts.
- **Main result.** A 10-percentage-point higher top-10 GSE lender share corresponds to 2.3 percentage points higher small-bank GSE growth; after controlling for the county's initial GSE share, the effect rises to 4.3 percentage points.
- **Economic meaning.** When the GSE market was leveled, small banks expanded GSE lending especially where the old large-lender advantage had been strongest.

3.10 Level-shock regressions for balance-sheet lending

The paper next studies whether the higher overall level of g-fees encouraged small banks to retain more loans on balance sheet.

- Areas with higher preexisting GSE activity should be more exposed to an increase in the cost of doing business with the GSEs.
- The balance-sheet results show that a 10-percentage-point higher GSE Share₂₀₁₁ corresponds to about 5.2 percentage points higher small-bank balance-sheet growth, and areas more dominated by top GSE lenders also see higher small-bank balance-sheet growth.
- **Interpretation.** When securitization becomes more expensive, retained lending becomes relatively more attractive. Small banks can exploit that margin; nonbanks largely cannot.

3.11 Validation exercises for the GSE-fee channel

- **Small nonbanks.** If the parity shock really reflects a GSE-segment funding shock, small nonbanks should also expand GSE lending in counties more exposed to the parity change. The paper finds exactly that.
- **Jumbo placebo.** The g-fee changes should not matter for jumbo loans, because those loans cannot be sold to the GSEs. The paper finds no relation between the g-fee shock variables and small-bank jumbo-loan growth.
- **Interpretation.** These validation tests strengthen the case that the measured effect really runs through GSE pricing rather than through unrelated county characteristics.

3.12 Within-bank and across-bank reallocation

The appendix studies whether small-bank substitution happens within individual banks' footprints, across different small banks, or both.

- **Within-bank extensive margin.** A 10-percentage-point lower Big4 growth makes a small bank about 1.5 percentage points more likely to enter a county.
- **Within-bank intensive margin.** Within a small bank's existing footprint, a 10-point lower Big4 growth corresponds to roughly a 2-point higher probability of positive small-bank growth in that county.
- **Across banks.** Small banks whose overall lending footprints were more exposed to Big4 retreat grew more at the lender level.
- **Interpretation.** The substitution pattern reflects both local reallocation by individual banks and differences in exposure across small banks, with the latter appearing quantitatively more important.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- If one simply regresses small-bank growth on Big4 growth, local demand shifts could contaminate the supply-side interpretation.
- Counties with a large postcrisis decline in Big4 lending may also have had changing borrower composition, local income shocks, or housing-market shocks that independently affect other lenders.
- This is why the paper places so much emphasis on county controls, state fixed effects, and instruments based on predetermined exposure.

4.2 Why the initial Big4 share helps identification

- The 2008 Big4 share is predetermined relative to postcrisis county-level lending changes.
- It captures the *scope for withdrawal*: where the Big4 started bigger, they could retreat more after the regulatory shock.
- Because the instrument uses pre-shock local exposure, it is less likely to be driven by the post-2008 demand shocks that would bias OLS.
- The paper further strengthens this claim by reestimating the IV with 2005 and 2002 Big4 shares, finding similar results.

4.3 Why lawsuit exposure helps identification

- The largest mortgage-related lawsuits were nationwide, bank-level shocks that fell primarily on the Big4.
- The exclusion restriction is more credible here because the shock originates from regulatory/legal exposure rather than from county-specific local conditions.
- Small banks were virtually untouched by these fines, so the lawsuit instrument mainly shifts the supply of the Big4 rather than the supply of small banks directly.

4.4 Additional credibility checks

- **Within-Big4 variation.** Bank of America was the Big4 member hit hardest by lawsuits and cut originations by roughly two-thirds. Small-bank growth was stronger in places where BoA retreated more, even after controlling for ex ante Big4 share.
- **Top-50 GSE robustness.** The GSE fee results are robust to measuring exposure using the top 50 rather than top 10 GSE sellers.
- **Placebo on jumbo loans.** The GSE shock variables do not predict small-bank growth in jumbo lending, as they should not if the mechanism truly runs through GSE pricing.

- **Decomposition of top GSE lenders.** The effect of non-Big4 top GSE lenders, especially nonbank top sellers, is also positive. This shows the GSE channel is distinct from the Big4 regulatory channel.

4.5 Relation to the literature

- The paper complements the literature on shadow banking and fintech growth by showing that there was also important substitution *within traditional banking*, not only between banks and nonbanks.
- It relates to work on Dodd-Frank size thresholds by showing that regulatory incidence varied systematically with bank size and mattered for local mortgage supply.
- It also connects to work on balance-sheet versus originate-to-distribute lending by showing that retained lending is exactly where small banks have their largest substitution advantage.
- Finally, it adds to the GSE and mortgage-finance literature by showing that funding-cost changes can reshape lender composition, not just loan pricing.

5 Tables and figures

5.1 Figure 1: Aggregate lending shares over time

Read:

- This is the paper's opening stylized fact. Small banks' national market share rises only modestly, from 16% to 19%, but that stability is remarkable because the rest of traditional banking shrinks dramatically.
- The Big4 fall from 33% to 12%; large banks fall from 13% to 8%; intermediate banks fall from 11% to 6%.
- Shadow banks surge from about 20% to 37%, and fintech from about 1% to 10%.
- The figure therefore sets up the puzzle correctly: small banks do not win in the aggregate the way nonbanks do, but they also do not collapse the way the rest of banking does.

5.2 Figure 2: Differential incidence of regulatory shocks across banks

Read:

- Mortgage servicing rights as a share of tier 1 capital are much higher for the Big4 than for the rest of the banking system.
- The Big4 start with lower capital ratios and must raise them the most after the crisis.
- Mortgage-related fines are an order of magnitude larger for the Big4 than for large banks, even before scaling by assets; scaled by assets, the Big4 are still hit the hardest.
- This figure establishes the paper's first main mechanism: postcrisis regulatory burden was highly uneven and particularly severe for the largest banks.

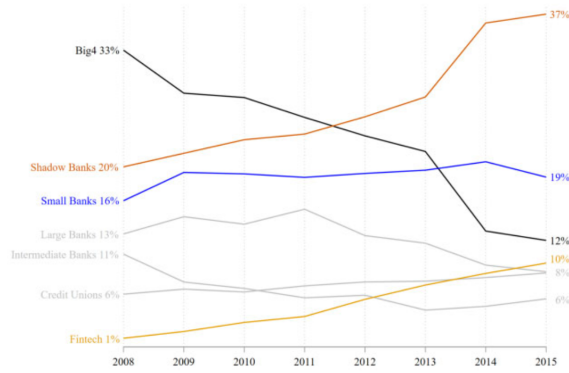


Figure 1
Aggregate lending shares over time
This figure presents the aggregate mortgage origination market shares for each lending class over time. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018).

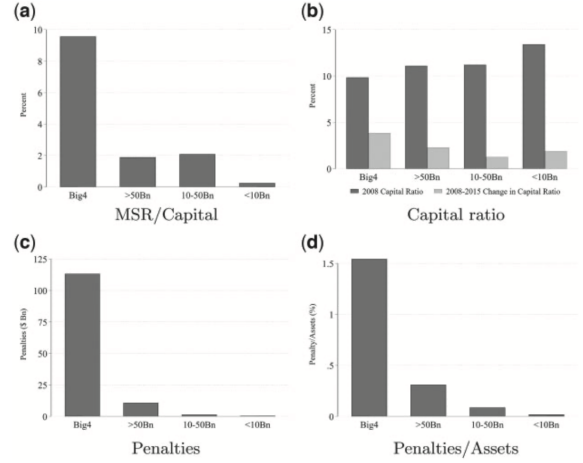


Figure 2
The Incidence of Regulatory Shocks Across Banks
This figure presents the degree to which banks of different sizes were exposed to postcrisis regulatory shocks (described in detail in Section 3). Panel a plots the mortgage servicing rights as a share of regulatory capital; panel b presents the 2008 tier1 capital and the changes from 2008 to 2015; panel c presents the raw magnitude of mortgage-related fines; and panel d presents those fines as a share of total assets. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; and *Small banks* includes banks with assets less than \$10 billion.

5.3 Table 2 and Figure 3: Big4 retreat and local lender substitution

Table 2
Lender class growth rate correlations

	Big4	Large banks	Interm banks	Small banks	Credit union	Shadow banks	Fintech
Big4	1.00						
Large banks	-0.10	1.00					
Intermediate banks	-0.03	0.08	1.00				
Small banks	-0.27	-0.01	-0.11	1.00			
Credit unions	-0.03	0.03	-0.08	0.05	1.00		
Shadow banks	-0.17	0.09	-0.05	0.07	0.09	1.00	
Fintech	-0.12	0.13	-0.00	0.17	0.06	0.27	1.00

This table presents the correlation matrix of county-level loan growth for each lender class from 2008 to 2015. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018). GSE loans are loans sold to Fannie Mae or Freddie Mac.

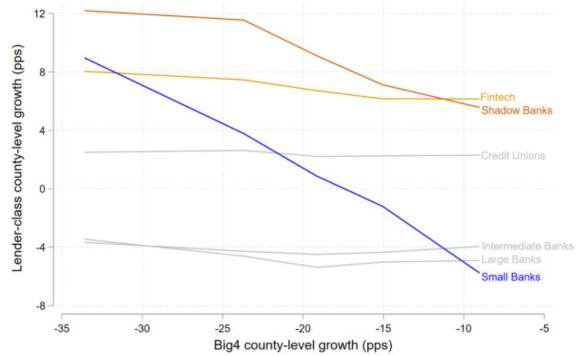


Figure 3
Big4 retreat and county-level lender class growth
This figure presents the county-level lender class growth from 2008 to 2015 as a function of the Big4 lending growth over the same period. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018).

Read:

- The county-level growth correlation between the Big4 and small banks is -0.27 , the most negative within-banking correlation.
- The Big4-small-bank relationship is more negative than the Big4-shadow-bank relationship (-0.17) and the Big4-fintech relationship (-0.12).
- Figure 3 shows a 14-percentage-point spread in small-bank growth between counties with the strongest and weakest Big4 retreat, versus only about 6 points for shadow banks.
- These objects tell the main descriptive story of the paper: locally, small banks are the clearest substitute for the Big4.

5.4 Table 4: OLS cross-county substitution estimates

Read:

- Small banks: coefficient -0.659 , so a 10-point lower Big4 growth is associated with 6.6 points higher small-bank growth.
- Shadow banks: -0.130 ; fintech: -0.047 . Those responses are much smaller.
- Large banks, intermediate banks, and credit unions show economically tiny responses.
- OLS already says that the most important local reshuffling is within traditional banking, from the Big4 to small banks.

Table 4
Cross-sectional heterogeneity in response to the Big4 retreat

	(1) Small banks	(2) Shadow banks	(3) Fintech	(4) Large banks	(5) Interm banks	(6) Credit unions
$Growth_{county}^{Big4}$	-0.659^{***} (<0.01)	-0.130^{***} (<0.01)	-0.047^{***} (<0.01)	-0.040^{***} (0.01)	0.002 (0.91)	-0.010 (0.41)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,983	2,983	2,983	2,983
R^2	.295	.418	.232	.309	.258	.224

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents OLS estimates from the regressions of county-level growth in mortgage originations for each lender class from 2008 to 2015 on the county-level growth in Big4 originations during this time period ($Growth_{county}^{Big4}$). *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018); *Large banks* includes banks with assets between \$50 billion and \$1 trillion; and *Intermediate Banks* includes banks with assets between \$10 billion and \$50 billion. Control variables include 2008 county-level logarithm of the 2008 population, non-white share of the population, per capita income, median age, share of the population with FICO below 660, county banking HHI based on deposits, house price growth, the logarithm of the number of banks with a physical branch in the county, the logarithm of the number of banks with at least one mortgage loan in the county, and the logarithm of the number of shadow banks or fintech lenders with at least one mortgage loan in the county. All regressions include state fixed effects. Standard errors are clustered by MSA.

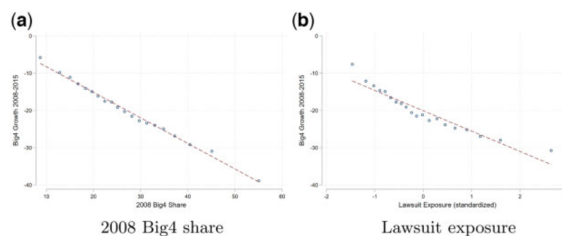


Figure 4
Instruments for Big4 growth during 2008–2015

This figure presents a bin scatter plot of county-level growth for the Big4 from 2008 to 2015 (y-axis) against the 2008 Big4 county share of mortgage originations (panel a) and standardized lawsuit exposure (panel b). *Big4* includes Bank of America, Citi, JP Morgan, and Wells Fargo.

5.5 Figure 4, Figure 5, and Table 5: IV evidence on the Big4 retreat

Read:

- Figure 4a shows the strength of the initial-share instrument: a 10-point higher 2008 Big4 share predicts 6.9 points lower Big4 growth from 2008 to 2015.
- Table 5 then shows that the IV effect for small banks is -0.884 with the 2008-share instrument and -0.970 with the lawsuit instrument.
- These imply that a 10-point bigger Big4 decline causes roughly 8.8 to 9.7 points higher small-bank growth, compared with only 2.6 to 3.5 points for shadow banks and 1.2 to 1.7 for fintech.
- The map in Figure 5 helps visualize the identifying variation: there is broad national variation both in initial Big4 presence and in subsequent Big4 retreat.

5.6 Table 6: Balance-sheet lending as the key channel

Read:

- For small banks, the non-balance-sheet substitution coefficient is -0.567 and the incremental balance-sheet effect is -0.580 , giving an overall balance-sheet response of -1.147 .

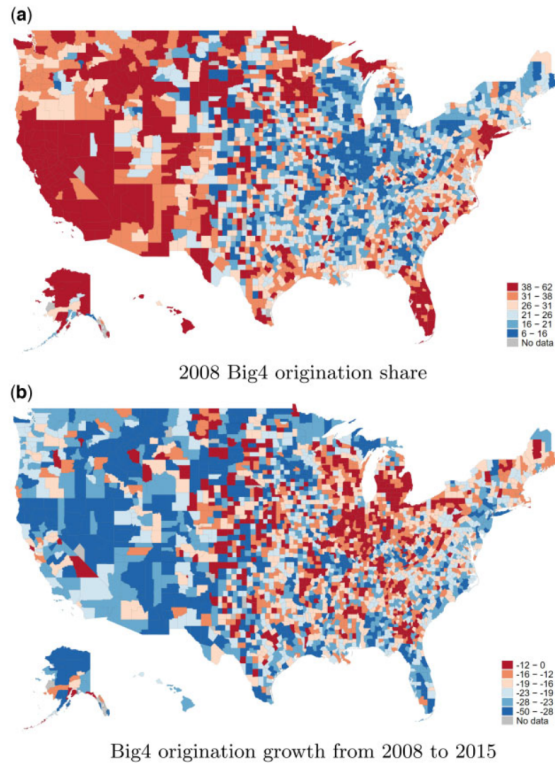


Figure 5
Mapping 2008 Big4 share and 2008–2015 Big4 growth
This figure presents the share of mortgages by the Big4 banks (Bank of America, Citi, JP Morgan, and Wells Fargo) in 2008 in panel a. Panel b plots the growth in the Big4 originations from 2008 to 2015.

Table 6
Substitution in balance sheet lending

	(1) Small banks	(2) Large banks	(3) Intermediate banks	(4) Shadow banks	(5) Fintech	(6) Credit Unions
$Growth_{county}^{Big4}$	−0.567*** (< 0.01)	−0.104*** (< 0.01)	−0.012 (0.64)	−0.375*** (< 0.01)	−0.123*** (< 0.01)	0.016 (0.56)
$Growth_{county}^{Big4} \times I[BS\text{-lending}]$	−0.580*** (< 0.01)	0.063 (0.10)	−0.002 (0.95)	0.422*** (< 0.01)	0.100*** (< 0.01)	−0.065 (0.13)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,954	5,954	5,954	5,954	5,954	5,954
R^2	.088	.019	.015	.061	.023	.014

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents IV estimates from the regressions of county-level growth in balance sheet or non-balance-sheet mortgage originations for respective lender classes from 2008 to 2015 on the county-level growth in Big4 originations during this time period. Each county has two observations: one for the lender class's growth in balance sheet lending (indicated by $I[BS\text{-lending}]$) and one for their growth in non-balance-sheet lending (the omitted category). $Growth_{county}^{Big4}$ represents the 2008–2015 loan growth for the Big4. The instruments for $Growth_{county}^{Big4}$ and $Growth_{county}^{Big4} \times I[BS\text{-lending}]$ are the 2008 county-level share of the Big4 and its interaction with $I[BS\text{-lending}]$. *Big4* represents Bank of America, Citi, JP Morgan, and Wells Fargo; *Small banks* includes banks with assets less than \$10 billion; *Large banks* includes banks with assets between \$50 billion and \$1 trillion; *Intermediate banks* includes banks with assets between \$10 billion and \$50 billion; *Shadow banks* include all nonbank, non-credit-union, and nonfintech lenders (such as independent mortgage companies); and *Fintech* includes nonbanks with a strong online presence and if nearly all of the mortgage application process takes place online with no human involvement from the lender (Buchak et al. 2018). Control variables are the same as those used in Table 4, and the regressions also include state $\times$ loan segment (balance sheet vs. non-balance-sheet) fixed effects. The Cragg-Donald F-statistic is 602. Standard errors are clustered by MSA.

- This means small-bank balance-sheet lending is roughly twice as responsive as non-balance-sheet lending.
- Shadow banks and fintech show the opposite pattern: their response is concentrated in non-balance-sheet lending, and their overall balance-sheet sensitivity is near zero.
- This table is arguably the most important mechanism table in the paper because it shows why small banks can out-substitute even lightly regulated nonbanks in local mortgage markets.

5.7 Figure 6 and Figure 7: GSE fee parity and the changing GSE market

Read:

- Figure 6 shows the two GSE fee shocks clearly: the gap between top lenders and small lenders closes, and average fees roughly double from about 30 to 60 basis points.
- The top-10 GSE lenders pay around 24 bps in 2008 versus around 33 bps for lenders outside the top 100; by 2013–2015 both groups are around 51–59 bps.
- Figure 7 shows the market consequence: the distribution of GSE lending becomes much flatter between 2011 and 2015, with less concentration among top lenders.
- Together, these figures show why the paper treats GSE funding-cost changes as a distinct supply-side shock rather than as merely another expression of postcrisis bank regulation.

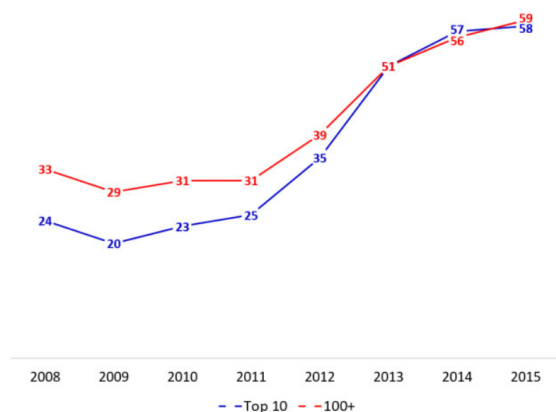


Figure 6
Change in the parity and levels of guarantee fees
This figure presents the average level of guarantee fees (in basis points) paid by lenders selling loans to the GSEs (Fannie Mae and Freddie Mac) over 2008–2015. We separately plot the g-fees paid by the top-10 GSE lenders and the g-fees paid by lenders outside the top-100 lenders.

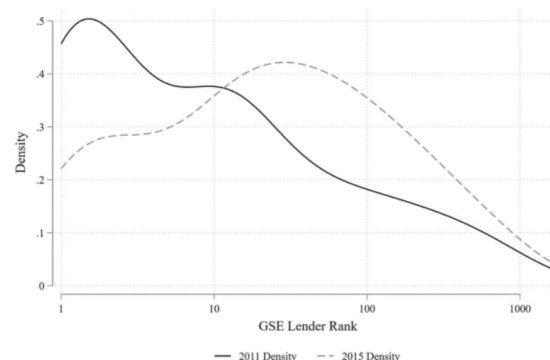


Figure 7
Changing distribution of GSE lending
This figure presents the density of GSE lending according to the rank of the lender. The GSE Lender Rank is computed annually by the total number of loans sold to the GSEs (which includes originations and loans purchased and subsequently sold to the GSEs within the year). We present the densities for 2011 and 2015.

5.8 Table 7: GSE-fee parity and level shocks

Read:

- In the GSE segment, a 10-point higher top-10 GSE lender share in 2011 predicts 2.3 points higher small-bank GSE growth, rising to 4.3 points after controlling for the county's initial GSE share.
- In balance-sheet lending, the strongest growth is in counties with higher prior GSE dependence and stronger exposure to the top GSE sellers, consistent with the level increase in g-fees making

retained lending more attractive.

- Small nonbanks also grow more in GSE lending in counties more exposed to the parity shock, which is exactly what the mechanism predicts.
- Jumbo lending shows no effect, providing a useful placebo test.

Table 7
Shocks to the costs of GSE lending and small bank growth

	Parity		Level		Validation	
	(1)	(2)	(3)	(4)	(5)	(6)
Lender class:	Small bank	Small bank	Small bank	Small bank	Small nonbank	Small bank
Loan Segment:	GSE	GSE	balance sheet	balance sheet	GSE	jumbo
Top10GSElenderShare2011	0.230*** (< 0.01)	0.429*** (< 0.01)		0.524*** (< 0.01)	0.128** (0.04)	0.098 (0.65)
GSE share 2011		-0.729*** (< 0.01)	0.515*** (< 0.01)	0.344*** (< 0.01)	-0.458*** (< 0.01)	0.120 (0.71)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,966	2,966	2,973	2,973	2,966	1,748
R ²	.020	.103	.035	.061	.127	.012

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents estimates from the regressions of county-level small bank lending growth in mortgage originations from 2011 to 2015 on measures of exposure to the g-fee parity and level shocks. The dependent variables are small bank growth in GSE lending (columns 1 and 2), small bank growth in balance sheet lending (columns 3 and 4), and small bank growth in jumbo lending (column 6). The dependent variable in column 5 is small nonbank growth in GSE lending, where small nonbanks are defined as nonbanks outside of the top-10 volume lenders. *Top10GSElenderShare2011* represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year. *GSE Share 2011* represents the share of all county loans in 2011 that were sold to the GSEs that year. Control variables are the same as those in Table 4 and include state fixed effects. Standard errors are clustered by MSA.

Table 8
Distinct role of the shocks to GSE funding

	All		GSE		Balance sheet	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>A. Top-10 GSE lender shares and small bank lending growth</i>						
Top10GSElenderShare2011	0.519*** (< 0.01)		0.429*** (< 0.01)		0.524*** (< 0.01)	
Top10Share Big4		0.565*** (< 0.01)		0.504*** (< 0.01)		0.575*** (< 0.01)
Top10Share nonbank		1.294*** (< 0.01)		0.733** (0.02)		1.256*** (0.01)
Top10Share other bank		0.450*** (< 0.01)		0.324*** (0.01)		0.441*** (< 0.01)
GSE share 2011	0.026 (0.70)	0.011 (0.87)	-0.729*** (< 0.01)	-0.744*** (< 0.01)	0.344*** (< 0.01)	0.328*** (< 0.01)
Observations	Yes	Yes	Yes	Yes	Yes	Yes
	2,983	2,983	2,966	2,966	2,973	2,973
R ²	.103	.113	.103	.106	.061	.064
<i>B. Top-50 GSE lender shares and small bank lending growth</i>						
Top50GSElenderShare2011	0.497*** (< 0.01)		0.425*** (< 0.01)		0.548*** (< 0.01)	
Top50Share Big4		0.575*** (< 0.01)		0.514*** (< 0.01)		0.594*** (< 0.01)
Top50Share nonbank		0.680*** (< 0.01)		0.741*** (< 0.01)		0.770*** (< 0.01)
Top50Share other bank		0.420*** (< 0.01)		0.310*** (< 0.01)		0.465*** (< 0.01)
GSE Share 2011	-0.047 (0.47)	-0.056 (0.39)	-0.797*** (< 0.01)	-0.809*** (< 0.01)	0.249** (0.02)	0.249** (0.03)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,983	2,983	2,966	2,966	2,973	2,973
R ²	.135	.140	.120	.127	.077	.077

p-values are in parentheses.

* $p < .10$; ** $p < .05$; *** $p < .01$.

This table presents estimates from the regressions of small banks' county-level lending growth in mortgage originations from 2011 to 2015 on measures of exposure to the g-fee cost shocks. The dependent variables are small bank lending growth for all loans (columns 1 and 2), GSE loans (columns 3 and 4), and balance sheet loans (columns 5 and 6). Panel A (B) uses the top-10 (top-50) GSE lenders to measure the degree of g-fee shock. *Top10GSElenderShare2011* represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year. *Top10Share X* represents the share of county lending in 2011 originated by lenders ranked in the top-10 sellers to the GSEs that year that are members of group X, where X is Big4, nonbank lenders, or non-Big4 banks. *GSE Share 2011* represents the share of all county loans in 2011 that were sold to the GSEs that year. The variables in panel B are defined analogously using the top-50 GSE lenders. Control variables are the same as those used in Table 4 and include state fixed effects. Standard errors are clustered by MSA.

5.9 Table 8: Distinct role of the GSE-funding channel

Read:

- The effect of top non-Big4 GSE lenders on small-bank growth remains positive and important, especially for balance-sheet lending.
- For top-10 lenders, the coefficient on top-10 nonbank GSE share is especially large in the balance-sheet specification.
- The same qualitative conclusion survives when the paper replaces the top-10 group with the top-50 group.
- This table matters because it addresses the obvious concern that GSE exposure is just repackaging Big4 exposure. The result is that the GSE funding-cost channel is genuinely distinct.

5.10 Appendix evidence: within-Big4 and within-small-bank reallocation

Read:

- Bank of America was the hardest-hit Big4 member, accounting for over half of the total fines and cutting origination by roughly two-thirds. Conditional on ex ante Big4 share, a 10-point lower BoA growth predicts about 3 points higher small-bank growth.
- Within a given small bank’s footprint, lower Big4 growth predicts more entry and a higher probability of positive growth. A 10-point lower Big4 growth raises county entry probability by about 1.5 points and raises the chance of positive growth by about 2 points.
- Across small banks, those with footprints more exposed to Big4 retreat grow more overall.
- These appendix results reinforce the paper’s interpretation that the small-bank response is not a mechanical artifact of aggregate shares; it is visible within banks, across banks, and within the Big4 itself.

6 Short summary

- **Main message.** Small banks persisted in the postcrisis mortgage market because two supply-side shocks favored them: regulation fell disproportionately on the Big4, and GSE fee changes opened room for smaller lenders.
- **Main mechanism.** Their distinctive advantage over nonbanks is the ability to lend on balance sheet. That is where their substitution for the Big4 is strongest.
- **Main empirical lesson.** Looking only at aggregate national shares misses the key margin of adjustment. The important reshuffling occurred locally, and much of it was a reallocation *within traditional banking* rather than simply a transfer from banks to nonbanks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that small banks were not a sideshow in the postcrisis mortgage market.
- Their endurance is explained by identifiable supply-side forces rather than by inertia or residual local monopoly power.
- Differential postcrisis regulation caused the Big4 to retreat, and small banks substituted for them especially strongly in balance-sheet lending.
- Changes in GSE guarantee-fee schedules created a second, distinct opportunity for small banks to expand in GSE lending and in retained lending.

7.2 Economic intuition

- The paper’s economic logic is simple but powerful: when the largest banks retreat because regulation becomes especially costly for them, the lenders best positioned to replace them are not necessarily the least regulated lenders, but the lenders with the right business model.
- Small banks are exactly such lenders. They are more lightly regulated than the Big4, but unlike nonbanks they can also retain loans on their balance sheet.

- When GSE securitization becomes more expensive, that balance-sheet option becomes even more valuable.
- So small banks win not because they dominate every segment, but because they are uniquely positioned at the intersection of local presence, lighter regulation, and retained-lending capacity.

7.3 Takeaway

This paper should change how one reads the post-2008 mortgage market. The standard narrative is “banks retreated, nonbanks replaced them.” Begley and Srinivasan show that this misses a crucial layer of heterogeneity. The Big4 retreated the most because regulation hit them the hardest. Small banks—not just shadow banks or fintech—stepped in where that retreat was strongest, and they did so through the lending margin that nonbanks are least able to serve: balance-sheet lending. The paper’s broader lesson is that regulatory and funding-cost shocks reshape credit supply through detailed differences in lender business models, not only through the broad bank-versus-nonbank split.